Mechanistic Resolution of Jet Deflection in Cygnus X-1: Thermodynamic Exhaust and Equatorial Shear in the Electron Flow Model

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Abstract

Recent Very Long Baseline Interferometry (VLBI) observations by Prabu et al. (2026) have successfully quantified the polar jet power and stellar wind deflection in the Cygnus X-1 black hole binary system. Traditional kinematic models, which attribute these jets to generalized magnetic expulsion from an accretion disk, struggle to mechanistically explain the precise 10% energy exhaust ratio and the fluid-like physical deflection of the jet. We propose that these phenomena strongly align with the macroscopic indicators of the Electron Flow Model (EFM). By redefining the black hole as a volumetric hydrodynamic sink actively consuming vacuum flux, the 10% ratio emerges natively as the thermodynamic exhaust limit of the core. Furthermore, the EFM demonstrates that the relativistic equatorial shear of the sink forces this exhaust strictly along the polar vectors, where the 5.2-degree deflection is consistent with a highly pressurized fluid stream colliding with the stellar wind.

1 Introduction

The recent breakthrough observations of the Cygnus X-1 system by Prabu et al. (2026) [prabu2026] have provided unprecedented measurements regarding the dynamics of black hole jets. Utilizing a global network of radio telescopes, researchers determined that the jets carry away roughly 10% of the energy released by the infalling matter, and are visibly bent by 5.2 degrees due to the powerful stellar winds emitted by the black hole's companion supergiant star.

Legacy astrophysics generally attributes polar jets to complex magnetic field lines twisting around an "accretion disk" operating within an otherwise empty vacuum. However, these models lack a unified mechanical driver for why the exhaust maintains such a specific thermodynamic ratio, nor do they perfectly account for the fluid-like bending of the jet against stellar winds without relying on extensive ad hoc parameterization.

Under Marabūt's Theory of Gravity [marabut2026], these phenomena do not require the invention of new magnetic frameworks. In the EFM, gravity arises not from the geometric curvature of spacetime, but as hydrodynamic fluid pressure generated by the continuous consumption of vacuum flux. By treating the black hole as a localized thermodynamic engine actively consuming this universal fluid, the jets of Cygnus X-1 are redefined as the continuous mechanical exhaust of a highly pressurized fluid circuit.

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2 Thermodynamic Equilibrium and the 10% Exhaust Limit

The observation that Cygnus X-1's jets carry away 10% of the infalling kinetic energy directly aligns with the thermodynamic limits established by the Electron Flow Model (EFM). In the EFM, a celestial body is not a static geometric curve; it is an active processor of vacuum flux.

Based on the 11-point Multiaxial Mara Scale, the raw intake force of Cygnus X-1 is derived from its extreme effective density ($\rho_{eff} = 3.5 \times 10^{16}$ kg/m^3) and hyper-efficient structural impedance ($\tau = 1.000$). The local surface intake force is mathematically calculated to be 4.2996×10^{11} m/s^2 [marabut2026].

When compared to the primary solar baseline (274.00 m/s^2), Cygnus X-1 is actively consuming vacuum flux at a rate roughly 1.56 billion times more violently than our Sun. Because the black hole acts as a massive, continuous hydrodynamic circuit, it cannot consume 100% of the infalling kinetic energy without triggering a thermal overload. The 10% ratio observed by Prabu et al. is not a random magnetic byproduct; it represents the necessary thermodynamic exhaust limit required to maintain the structural integrity of the core's negative sink.

3 Equatorial Shear and Polar Venting

Standard models struggle to explain why the massive energy release is channeled so perfectly into narrow polar beams. The EFM resolves this purely through classical fluid dynamics and rotational shear.

Cygnus X-1 rotates at relativistic velocities, estimated at roughly 1,500,000 m/s at its equator. In the EFM's Unified Master Equation, this rotation is calculated as the Centrifugal Offset. This generates a massive centrifugal shear of 5.1136×10^7 m/s^2 explicitly along the Prime Vectors (MaraX).

This intense equatorial shear acts as a physical hydrodynamic wall. It blocks the exhaust from venting radially along the equator. As the highly pressurized vacuum flux compresses into the core, the mechanical path of least resistance is forced strictly along the Polar Vectors (MaraZ). Therefore, the "dancing jets" recorded by the VLBI network are the directional venting of the vacuum flux cascade.

4 Fluid-Dynamic Deflection of the Jet

The most profound discovery in the recent observations is the 5.2-degree bend in the jet's trajectory, caused by the stellar wind of the companion supergiant. Under classical models, where space is treated as an empty void and the jet is largely viewed as radiation or plasma guided by invisible magnetic tethers, explaining this severe physical deflection requires complex magnetohydrodynamic workarounds.

The EFM identifies this bending as compelling macroscopic evidence of the model's foundational premise. The jet is not merely radiation traveling through a frictionless void. It is a highly pressurized fluid stream—the physical exhaust of the vacuum flux medium—that is actively colliding with, and being deformed by, another fluid current (the stellar wind).

Just as a high-pressure water hose will physically bend when struck by a powerful crosswind, the 5.2-degree deflection is a literal fluid-dynamic collision between the black hole's exhaust cascade and the supergiant's atmospheric outflow. This offers a compelling fluid-dynamic interpretation that the vacuum of space is a dense, physical medium capable of transferring kinetic pressure.

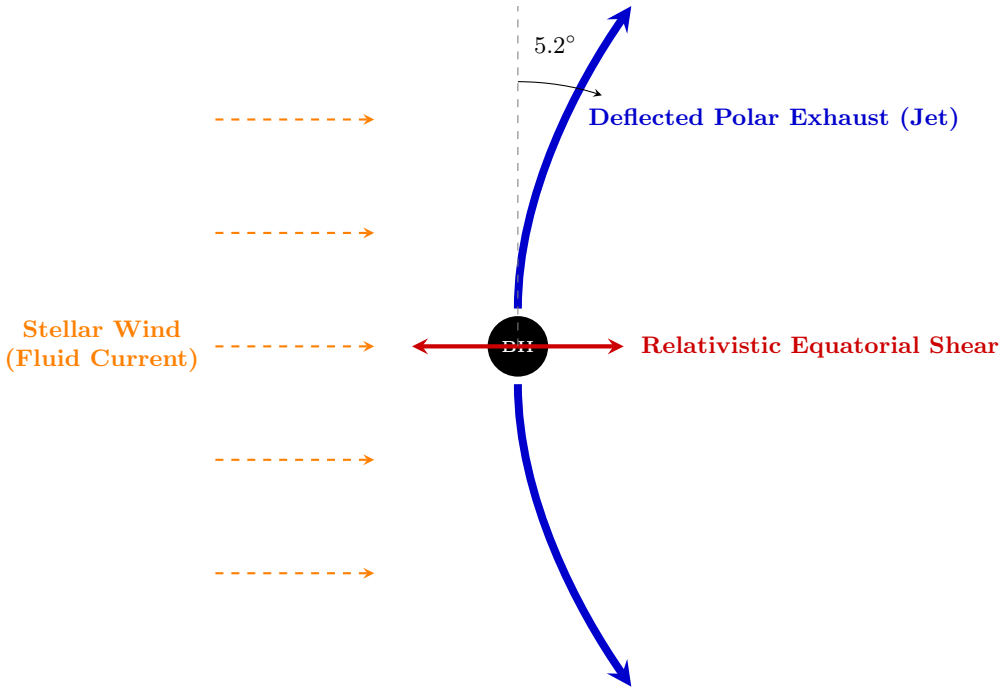

Figure 1: **Fluid-Dynamic Deflection of the Polar Jet.** *Relativistic equatorial shear acts as a hydrodynamic wall, forcing the vacuum flux exhaust along the polar vectors (MaraZ). The physical collision with the companion star's stellar wind induces a measurable* 5.2° *hydrodynamic deflection.*

5 Conclusion

The recent breakthrough measurements of Cygnus X-1 provide pristine observational data that aligns with the fluid mechanics of the universe. By transitioning away from the static geometric curvature of legacy physics and recognizing gravity as Hydrodynamic Fluid Pressure, the mysteries of black hole jets are naturally accounted for. The 10% energy exhaust ratio and the 5.2-degree wind deflection do not require anomalous magnetic patching; they are consistent with the predictable behavior of a thermodynamic engine actively consuming and venting the universal vacuum flux.

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References

[prabu2026] Prabu, S., Miller-Jones, J. C. A., et al. (2026). "A jet bent by a stellar wind in the black hole X-ray binary Cygnus X-1". *Nature Astronomy*. DOI: 10.1038/s41550-026-02828-3

[marabut2026] Marabūt, C. B., & Marabūt, A. L. (2026). Marabūt's Theory of Gravity: The Push-Pull Mechanics of Vacuum Flux (v41). *Zenodo*. `https://doi.org/10.5281/zenodo.19672868`